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Unit 6

GASEOUS STATE

So many of the properties of properties of matter, especially when in gaseous form, can be deduced from hypothesis that their minute parts are in rapid motion, the velocity increasing with temperature, that the precise nature of this motion becomes a subject of rational curiosity.

James Clerk Maxwell





Jacques Charles
(1746-1823)

Learning Objectives

After studying this unit, students will be able to

- State the laws governing an ideal gas
- Demonstrate gas laws in various real life situations
- Derive ideal gas equation and perform calculations using it.
- State the deviations from ideal behavior
- Derive van der Waals equation.
- Define Graham's law of diffusion
- Define compressibility factor
- Explain critical phenomena
- Derive critical constants in terms of van der Walls constants
- Explain Andrew's isotherms of carbon dioxide
- Describe Joule – Thomson effect and liquefaction of gases.

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